UD 0. Comandos de red en linux

Por cada apartado captura pantalla de que del comando que se ejecuta y de la salida que se ha ejecutado correctamente.

1. Ejercicio: Ver las tarjetas de red físicas en Ubuntu Server

Objetivo: Identificar las tarjetas de red físicas del equipo, diferenciarlas de las interfaces virtuales (lo, docker0, veth, bridges, etc.) y obtener información del driver y del hardware.

Nota: Los nombres de las interfaces físicas suelen ser eth0, enp0s3, ens33, eno1, etc. Las virtuales son lo, docker0, br-*, veth*, virbr0, etc.

Comandos resumen a utilizar

- Ver todas las interfaces — ip -br link show
- Ver solo hardware PCI de red — lspci | grep -i ethernet
- Ver hardware de red resumido — sudo lshw -class network -short
- Ver detección del kernel — dmesg | grep -i eth
- Ver driver cargado — lsmod | grep -i <driver>
- Ver driver de una interfaz — sudo ethtool -i <interfaz>

1.1. Listar todas las interfaces de red (físicas y virtuales).

```
alumno@asir2-07:~$ ip -br link show
lo               UNKNOWN        00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
enp128s31f6      UP             f0:4e:a4:39:03:32 <BROADCAST,MULTICAST,UP,LOWER_UP>
docker0          DOWN           9a:50:b5:38:41:dd <NO-CARRIER,BROADCAST,MULTICAST,UP>
vboxnet0         UP             0a:00:27:00:00:00 <BROADCAST,MULTICAST,UP,LOWER_UP>
vboxnet1         DOWN           0a:00:27:00:00:01 <BROADCAST,MULTICAST>
vboxnet10        DOWN           0a:00:27:00:00:0a <BROADCAST,MULTICAST>
vboxnet11        DOWN           0a:00:27:00:00:0b <BROADCAST,MULTICAST>
vboxnet12        DOWN           0a:00:27:00:00:0c <BROADCAST,MULTICAST>
vboxnet2         DOWN           0a:00:27:00:00:02 <BROADCAST,MULTICAST>
vboxnet3         DOWN           0a:00:27:00:00:03 <BROADCAST,MULTICAST>
vboxnet4         DOWN           0a:00:27:00:00:04 <BROADCAST,MULTICAST>
vboxnet5         DOWN           0a:00:27:00:00:05 <BROADCAST,MULTICAST>
vboxnet6         DOWN           0a:00:27:00:00:06 <BROADCAST,MULTICAST>
vboxnet7         DOWN           0a:00:27:00:00:07 <BROADCAST,MULTICAST>
vboxnet8         DOWN           0a:00:27:00:00:08 <BROADCAST,MULTICAST>
vboxnet9         DOWN           0a:00:27:00:00:09 <BROADCAST,MULTICAST>
```

1.2. Ver solo las interfaces que tienen un dispositivo PCI físico asociado

```
alumno@asir2-07:~$ lspci | grep -i ethernet
80:1f.6 Ethernet controller: Intel Corporation Device 550c (rev 10)
```

1.3. Obtener información detallada del hardware de red

```
alumno@asir2-07:~$ lshw -class network -short
AVISO: debería ejecutar este programa como superusuario.
ruta H/W        Dispositivo   Clase          Descripción
=====================================================
/0/1f.6         enp128s31f6   network        Intel Corporation
AVISO: la salida puede ser incompleta o imprecisa, debería ejecutar este programa como superusuario.
```

1.4. Ver los mensajes del kernel al detectar las tarjetas

```
alumno@asir2-07:~$ dmesg | grep -i eth
[    1.496932] e1000e 0000:80:1f.6 eth0: (PCI Express:2.5GT/s:Width x1) f0:4e:a4:39:03:32
[    1.496937] e1000e 0000:80:1f.6 eth0: Intel(R) PRO/1000 Network Connection
[    1.497131] e1000e 0000:80:1f.6 eth0: MAC: 15, PHY: 12, PBA No: FFFFFF-0FF
[    1.676788] e1000e 0000:80:1f.6 enp128s31f6: renamed from eth0
```

1.5. Comprobar los módulos del kernel cargados para las tarjetas

```
alumno@asir2-07:~$ lsmod | grep -i e1000e
e1000e                360448  0
```

1.6. Ver el driver y detalles de una interfaz concreta

```
alumno@asir2-07:~$ ethtool -i enp128s31f6
driver: e1000e
version: 6.17.0-1025-oem
firmware-version: 0.5-4
expansion-rom-version:
bus-info: 0000:80:1f.6
supports-statistics: yes
supports-test: yes
supports-eeprom-access: yes
supports-register-dump: yes
supports-priv-flags: yes
```

Responde a estas preguntas:

¿Cuántas tarjetas de red físicas tiene el equipo?

- solo 1

¿Cuál es la dirección MAC de cada una?

- f0:4e:a4:39:03:32

- ip link show enp128s31f6

¿Qué driver utiliza cada tarjeta?

- driver: e1000e

- ethtool -i enp128s31f6

¿Qué interfaces son virtuales y por qué?

- 14

- una se usa para looback, otra para dockers, y las otras 12 son creadas por nosotros en virtualbox

2. Ejercicio : Ver todas las interfaces de red disponibles

Resumen de comandos a utilizar

- Ver interfaces ip link show / ip addr show
- Ver resumen ip -br link / ip -br addr
- Activar interfaz sudo ip link set <if> up
- Desactivar interfaz sudo ip link set <if> down
- Añadir IP sudo ip addr add <IP>/<prefijo> dev <if>
- Eliminar IP sudo ip addr del <IP>/<prefijo> dev <if>
- Ver tabla ARP ip neigh show

2.1. Muestra todas las interfaces con su estado:

```
alumno@asir2-07:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp128s31f6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether f0:4e:a4:39:03:32 brd ff:ff:ff:ff:ff:ff
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT group default
    link/ether 9a:50:b5:38:41:dd brd ff:ff:ff:ff:ff:ff
4: vboxnet0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff
5: vboxnet1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:01 brd ff:ff:ff:ff:ff:ff
6: vboxnet10: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:0a brd ff:ff:ff:ff:ff:ff
7: vboxnet11: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:0b brd ff:ff:ff:ff:ff:ff
8: vboxnet12: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:0c brd ff:ff:ff:ff:ff:ff
9: vboxnet2: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:02 brd ff:ff:ff:ff:ff:ff
10: vboxnet3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:03 brd ff:ff:ff:ff:ff:ff
11: vboxnet4: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:04 brd ff:ff:ff:ff:ff:ff
12: vboxnet5: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:05 brd ff:ff:ff:ff:ff:ff
13: vboxnet6: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:06 brd ff:ff:ff:ff:ff:ff
14: vboxnet7: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:07 brd ff:ff:ff:ff:ff:ff
15: vboxnet8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:08 brd ff:ff:ff:ff:ff:ff
16: vboxnet9: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:09 brd ff:ff:ff:ff:ff:ff
```

2.2. Muestra solo las interfaces activas (UP):

```
alumno@asir2-07:~$ ip link show up
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp128s31f6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether f0:4e:a4:39:03:32 brd ff:ff:ff:ff:ff:ff
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT group default
    link/ether 9a:50:b5:38:41:dd brd ff:ff:ff:ff:ff:ff
4: vboxnet0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0a:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff
```

2.3. Muestra las direcciones IP de todas las interfaces:

```
alumno@asir2-07:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp128s31f6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether f0:4e:a4:39:03:32 brd ff:ff:ff:ff:ff:ff
    inet 172.31.7.1/16 brd 172.31.255.255 scope global noprefixroute enp128s31f6
       valid_lft forever preferred_lft forever
    inet 10.230.99.119/23 brd 10.230.99.255 scope global dynamic noprefixroute enp128s31f6
       valid_lft 2572sec preferred_lft 2572sec
    inet6 fe80::16fb:b1d9:f453:768d/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 9a:50:b5:38:41:dd brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
4: vboxnet0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0a:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.1/24 brd 192.168.56.255 scope global vboxnet0
       valid_lft forever preferred_lft forever
    inet6 fe80::800:27ff:fe00:0/64 scope link
       valid_lft forever preferred_lft forever
5: vboxnet1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:01 brd ff:ff:ff:ff:ff:ff
6: vboxnet10: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:0a brd ff:ff:ff:ff:ff:ff
7: vboxnet11: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:0b brd ff:ff:ff:ff:ff:ff
8: vboxnet12: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:0c brd ff:ff:ff:ff:ff:ff
9: vboxnet2: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:02 brd ff:ff:ff:ff:ff:ff
10: vboxnet3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:03 brd ff:ff:ff:ff:ff:ff
11: vboxnet4: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:04 brd ff:ff:ff:ff:ff:ff
12: vboxnet5: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:05 brd ff:ff:ff:ff:ff:ff
13: vboxnet6: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:06 brd ff:ff:ff:ff:ff:ff
14: vboxnet7: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:07 brd ff:ff:ff:ff:ff:ff
15: vboxnet8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:08 brd ff:ff:ff:ff:ff:ff
16: vboxnet9: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:09 brd ff:ff:ff:ff:ff:ff
```

3. Ejercicio : Desactivar una tarjeta de red

3.1. Comprueba que la interfaz está activa (ej. enp0s8):

```
alumno@Router1:~$ ip link show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
```

3.2. Desactívala:

```
alumno@Router1:~$ sudo ip link set enp0s8 down
[sudo] password for alumno:
alumno@Router1:~$ ip link show enp0s8
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
```

3.3. Verifica que ahora está DOWN:

```
alumno@Router1:~$ sudo ip link set enp0s8 down
[sudo] password for alumno:
alumno@Router1:~$ ip link show enp0s8
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
```

4. Ejercicio : Activar una interfaz de red

Objetivo: Poner en estado UP una interfaz que está DOWN.

4.1. Comprueba el estado actual de la interfaz (supongamos enp0s8):

```
alumno@Router1:~$ ip link show enp0s8
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
```

4.2. Actívala:

- sudo ip link set enp0s8 up

4.3. Verifica que ahora está UP:

```
alumno@Router1:~$ ip link show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
```

5. Configurar varias direcciones IP en una misma interfaz

Objetivo: Asignar dos direcciones IP adicionales a una interfaz (sin eliminar la que ya tiene).

5.1. Asegúrate de que la interfaz está activa (ej. enp0s8):

```
alumno@Router1:~$ ip addr show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.100.3/24 metric 100 brd 10.0.100.255 scope global dynamic enp0s8
       valid_lft 484sec preferred_lft 484sec
    inet6 fe80::a00:27ff:fe9a:2f9/64 scope link
       valid_lft forever preferred_lft forever
```

5.2. Añade una primera IP:

- sudo ip addr add 192.168.1.10/24 dev enp0s8

5.3. Añade una segunda IP:

- sudo ip addr add 192.168.1.20/24 dev enp0s8

5.4. Verifica que ambas están asignadas:

```
alumno@Router1:~$ sudo ip addr add 192.168.1.10/24 dev enp0s8
alumno@Router1:~$ ip addr show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.100.3/24 metric 100 brd 10.0.100.255 scope global dynamic enp0s8
       valid_lft 391sec preferred_lft 391sec
    inet 192.168.1.10/24 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe9a:2f9/64 scope link
       valid_lft forever preferred_lft forever
```

```
alumno@Router1:~$ sudo ip addr add 192.168.1.20/24 dev enp0s8
alumno@Router1:~$ ip addr show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.100.3/24 metric 100 brd 10.0.100.255 scope global dynamic enp0s8
       valid_lft 328sec preferred_lft 328sec
    inet 192.168.1.10/24 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet 192.168.1.20/24 scope global secondary enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe9a:2f9/64 scope link
       valid_lft forever preferred_lft forever
```

6. Ejercicio : Eliminar una dirección IP de una interfaz

Objetivo: Borrar una de las direcciones IP previamente asignadas.

6.1. Muestra las IPs actuales de la interfaz (ej. enp0s8):

```
alumno@Router1:~$ ip addr show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.100.3/24 metric 100 brd 10.0.100.255 scope global dynamic enp0s8
       valid_lft 536sec preferred_lft 536sec
    inet 192.168.1.10/24 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet 192.168.1.20/24 scope global secondary enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe9a:2f9/64 scope link
       valid_lft forever preferred_lft forever
```

6.2. Elimina una IP

- sudo ip addr del 192.168.1.10/24 dev enp0s8

6.3. Verifica que ya no está:

```
alumno@Router1:~$ sudo ip addr del 192.168.1.10/24 dev enp0s8
alumno@Router1:~$ ip addr show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9a:02:f9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.100.3/24 metric 100 brd 10.0.100.255 scope global dynamic enp0s8
       valid_lft 475sec preferred_lft 475sec
    inet 192.168.1.20/24 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe9a:2f9/64 scope link
       valid_lft forever preferred_lft forever
```

7. Ejercicio 6: Ver la tabla ARP (vecinos)

Objetivo: Visualizar las entradas ARP del sistema, tanto las aprendidas dinámicamente como las estáticas.

7.1. Muestra la tabla ARP actual:

```
alumno@Router1:~$ ip neigh show
10.0.100.1 dev enp0s8 lladdr 52:55:0a:00:64:01 STALE
10.0.100.2 dev enp0s8 lladdr 08:00:27:24:fe:3a STALE
192.168.56.1 dev enp0s3 lladdr 0a:00:27:00:00:00 REACHABLE
1.1.1.1 dev enp0s9 FAILED
8.8.8.8 dev enp0s9 FAILED
1.1.1.1 dev enp0s10 FAILED
8.8.8.8 dev enp0s10 FAILED
192.168.56.100 dev enp0s3 lladdr 08:00:27:c8:82:5e STALE
```